Orange County Fire Services
Operational Area Plan Annex

Section 9

Waste Water Discharge
Activities Plan

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Orange County Fire Services
Operational Plan

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INTENT

Purpose of this plan is to provide information as it relates to waste water discharge regulatory requirements. These requirements are part of the Municipal Storm-water permit process of the California Regional Water Quality Boards. Orange County falls under both San Diego and Santa Ana Regional Water Quality Boards.

The Orange County Storm water Permittees (Permittees), in cooperation with the Orange County Fire Chief's Association, have developed fire department activities procedures to provide guidance to firefighting personnel in performing activities that could impact the quality of urban and storm water runoff. Guidance is provided in the form of recommended Best Management Practices (BMP) that are shared and incorporated as part of the countywide Operational Annex Plan.

The Best Management Practices (BMP) when followed will minimize run-off releases associated with firefighting, fire training, emergency scene spills and discharges, trauma scene waste management and fire facility maintenance activities.

PROCEDURE

Non-emergency Firefighting Activities

1. Discharges Associated With Fire Training Activities

Training activities, which simulate emergency responses, must be performed in a manner that reduces or prevents discharges to the storm water systems whenever practicable. In addition, when the elimination of discharges into the storm water system is unavoidable (i.e. equipment failures), measures will be implemented to minimize impacts to water quality:

- Fire training should be conducted, where feasible, in facilities where runoff controls protecting the storm drain system have been engineered and built into the facility.

- Direct water flows to landscape or green belt areas whenever possible.

- When flowing water that cannot be contained in the landscaped areas, survey the area prior to the training exercise to ensure that debris will not enter the storm water system as a result of the flows generated during the drill.

- When practicable and necessary, divert flows to the sewer, with the permission of the local sewer agency.

- Use fog streams for short durations.

- Use lower gallon per minute (GPM) nozzle settings.
- Prevent discharge of foam to the storm drain system. If training activities involve the use of foam, block off all potentially affected storm drain inlets with plastic sheeting and sandbags or temporary berms, collect the runoff, and discharge to the sanitary sewer after a de-foaming agent has been applied.

2. Discharges Associated With Post-Emergency Fire Fighting Activities

The post-emergency rehabilitation and maintenance of response equipment must be performed in a manner that prevents discharges to the storm water system whenever practicable, and minimizes discharges to the storm water system when elimination of discharges is practically unavoidable.

3. Discharges Associated with Activities Conducted at Fire Facilities

Activities conducted at fire facilities include: vehicle and equipment washing and cleaning, vehicle fueling, vehicle and equipment maintenance and repair, hose washing and cleaning, facility maintenance, and solid and hazardous waste storage. It should be noted that the activities of vehicle fueling and vehicle, equipment maintenance and repair, and hazardous materials storage are also considered Municipal Activities. As such, the BMPs listed below for these activities have also been incorporated into the Municipal Activities Program element.

A. Vehicles and Equipment Washing and Cleaning

Reducing pollutant discharges from vehicle and equipment washing and cleaning involves both short and long-term solutions. In the short-term (prior to any significant station rehabilitation or repair), both low cost structural and permanent non-structural BMPs can be used such as placing spill kits at fueling stations and training personnel. For example, low cost structural BMPs may include installing storm drain inlet inserts that trap or remove sediment from trucks and/or reducing the frequency and duration of vehicle and equipment washing and cleaning activities that produce the pollutants or discharges. Long-term capital improvement projects, such as connecting permanent wash racks to the sanitary sewer, should also be considered if necessary and/or when the short-term use of BMPs is inefficient or ineffective.

The following BMPs should be considered in order to prevent or reduce the discharge of pollutants to the storm drain system from vehicle and equipment washing and cleaning:

- Use methods of cleaning vehicles that employ the minimal use of water, such as wet chamois or non-water rinses, when applicable;

- Limit the use of all cleaning agents and when feasible only use water.

- Remove debris from any area or facility used for washing and/or cleaning vehicles;

- Prevent runoff from vehicle and equipment washing and cleaning from entering the storm drain to the extent feasible by employing one of the following BMPs:

    - Direct water flows to landscaped or green areas or containing the water onsite and allow it to evaporate and infiltrate whenever safe to do so without causing damage or erosion;

    - Use designated wash areas (preferably covered and bermed) to contain and/or divert the wash water to the sewer either through the use of “wet-vac” or through a plumbed sanitary sewer connection;

    - Use self-contained water recycling systems;

    - Use off-site commercial washing and steam cleaning facilities.

- Prohibit all steam cleaning discharges from entering the storm drain. Direct all steam cleaning discharges to the sanitary sewer.

B. Vehicle Fueling

When fueling fire vehicles, the operators should prevent fuel spills and leaks, and reduce their impacts to storm water by using the following BMPs:

- Protect the fueling area from storm water by installing a canopy.

- Avoid fueling during active rain conditions at un-canopied fuel areas.

- When necessary to re-fuel during active rain conditions at non-canopied fueling areas, the person refueling shall not leave the fueling area while fuel is flowing from the pump.

- Pave fueling area surfaces with Portland cement concrete (or other equivalent smooth impervious surface) with a 2% to 4% slope to

prevent puddles.

- Keep perimeter drains clear of debris at all times;
- Where a perimeter drain is not installed, install a berm or grade area to prevent run-on of storm water and spilled liquids.
- Use a dead-end sump to collect spills or install an oil-water separator.
- Utilize vapor recovery nozzles to help control drips as well as air pollution.
- Discourage “topping-off” of fuel tanks.
- Maintain a spill control kit at the site. Use absorbent materials on small spills and general cleaning rather than hosing down an area. Remove the absorbent materials promptly and dispose as hazardous waste.
- Discourage the practice of “mobile fueling” during non-emergency events, if the vehicles and/or equipment can be brought to the refueling facility.

C. Vehicles and Equipment Maintenance and Repair

Prevent or reduce the discharge of pollutants to the storm drain from vehicle and equipment maintenance and repair by running a “dry shop” by employing the following BMP’s

- Conduct vehicle and equipment maintenance in areas where precautions have been taken to prevent the entry of spills into the storm drain system.
- Use dry cleaning methods in maintenance and repair areas unless adequate precautions have been taken to prevent the discharge of wash water to the storm drain.

D. Hose Washing and Cleaning

Prevent or reduce the discharge of pollutants to storm water from fire hose and equipment washing and cleaning by employing the following BMPs, as appropriate.

- Design new areas or facilities used for washing and/or cleaning fire hoses to prevent wash water or other debris from entering the storm drain or receiving water without adequate treatment.

- Prevent runoff from hose washing and cleaning from entering the storm drain to the extent feasible by employing one of the following BMPs:

    - Direct water flows to landscaped or green areas or containing the water onsite and allowing it to percolate through plant material, the landscape, or to evaporate completely, whenever safe to do so without causing damage or erosion;

    - Use designated wash areas (preferably covered and bermed) to contain and/or divert the wash water to the sewer either through the use of a “wet-vac” or through a plumbed sanitary sewer connection.
- Prevent wash water containing detergents, degreasers, or other contaminants from entering the storm drain.

- When cleaning the wash area prevent discharge from entering the storm drain. Utilize wet mop cleaning methods in small areas, when feasible.

- Use methods of cleaning fire hoses that employ the minimal use of water, such as high-pressure spray washers, when applicable.

- Consider the use of biodegradable cleaning agents.

E. Facility Maintenance

Facility maintenance activities conducted by either contract or in-house personnel should be performed using the following BMPs to prevent pollutants from entering the storm drain system:

- Use dry cleaning methods, such as sweeping, to clean impervious hardscape areas such as apparatus floors, driveways, patios, and walkways. Place sweepings and debris in receptacles for solid waste disposal.

- Maintain vegetated landscape areas as required to limit introduction of leaves and other landscape waste into the storm water system.

- Monitor and maintain irrigation systems as required to minimize off site discharge of irrigation runoff.

- Provide spill kits at the work site. Use spill kits for cleanup when appropriate. Dispose of spilled material in accordance with applicable regulations.

- Maintain and repair structures using methods that do not result in the release of water, soils, or waste to the storm drain system.

F. Solid Waste and Hazardous Materials Storage Areas

- Provide a canopy or roof for solid waste and hazardous materials storage areas;

- Provide secondary containment (i.e. a metal or plastic pan with a raised edge) for hazardous materials storage areas;

- Provide lids for all trash and solid waste receptacles. Keep lids closed to prevent contact with rainfall and to ensure containment of waste within the storage area.

Emergency Fire Fighting Activities

An “emergency” exists from alarm notification until, in the opinion of the incident commander, the emergency has concluded and emergency equipment is returned to station. Discharges occurring during emergency fire fighting activities (i.e. flows necessary for the protection of life and property) do not require BMPs and are not prohibited under the storm water permits. However, when and where possible and practicable, and when not interfering with health and safety, implementation of all applicable BMPs described this section should be considered.

1. Trauma Scene Cleanup

Trauma scene cleanup requires special procedures in addition to the general cleanup procedures described above. Trauma scene wastes (i.e. blood and human tissue) may be generated at various types of crime or accident scenes. These types of wastes can pose a serious human health risk to those who are responsible for overseeing and conducting the cleanup. It is important that the procedures described below be employed when feasible to protect personnel and prevent any unnecessary discharges of material into the storm drain. The types of blood borne pathogens that may be encountered at a trauma scene include, but are not limited to HIV, Hepatitis B and Hepatitis C.

A. General Trauma Scene Cleanup Procedures

- Management Act, *California Health and Safety Code, Sections 117600 – 118360*. The Act may be viewed on the California Department of Health Services web page at: http://www.cdph.ca.gov/certlic/medicalwaste/Documents/MedicalWaste/MedicalWasteManangementAct.pdf www.dhs./ca./gov/ps/ddwem/environmental/Med_Waste/mdm_act.pdf. Fire Departments on emergency calls who clean up a scene themselves are not required to have a Biowaste Hauling Permit to remove liquid or solid trauma scene waste(s) from the scene for temporary storage or disposal. The County of Orange Medical Waste Section (Emergency Response through Control 1 after hours at (714) 628-7008) is available for consultation/assistance with trauma scene management.

- A Registered Trauma Scene Management Practitioner (RTSMP) is necessary when a scene requires decontamination and cleanup beyond the capability of Public Agency personnel on scene. RTSMPs are registered with the Department of Public Health pursuant to Section 118321 of the Medical Waste Management Act. A list of RTSMP's can be obtained from the California Department of Health Services web page at: Trauma scene cleanup operations must be performed in accordance with the Medical Waste at: (www.cdph.ca.gov/certlic/medicalwaste/Documents/MedicalWaste/2013/Practitioners.pdf). Additional information may be obtained by calling the California Department of Public Health at 916-558-1784.

- Universal precautions and adequate personal protective equipment must be utilized during any clean-up operation. Additional information and guidance for cleanup policies and procedures for bodily fluids may be obtained 24 hours a day by contacting the Orange County Health Care Agency (HCA), Regulatory Health Services. During Business hours call (714) 667-3600; after hours, call Control One at (714) 628-7008.

- Implement appropriate BMPs and protect the storm drain system to the maximum extent practicable when conducting cleanup of trauma scenes. Notifications regarding significant potential impacts to the storm drain system and additional information and guidance on BMPs to be implemented may be obtained 24 hours a day by contacting the

County of Orange O.C. Public Works at (877) 897-7455 or through Control One at (714) 628-7008.

B. Cleanup of Trauma Scene Wastes on Public Property

Public Agency personnel shall observe the following points:

- Cleanup of trauma scene waste(s) on public property must be performed by properly trained (OSHA- blood borne pathogen trained) Public Agency personnel (police, fire or safety personnel), or by a Registered Trauma Scene Management Practitioner (RTSMP) called to the scene by the incident commander.

- Before proceeding with site cleanup, Public Agency personnel must possess proper personal protective equipment, adequate supplies of bleach and sufficient water to properly complete the cleanup operation.

- Wear gloves when washing or handling contaminated equipment, clothing or other materials. Wear other personal protective gear as necessary according to the potential for splashing.

- The Coroner's Office should remove any human tissue or body parts at the scene of a fatality. Coroner's Office personnel should follow their own departmental policies and procedures for removing any human tissue found on scene. The Coroner will not respond to a non-fatal incident for the purpose of human tissue removal.

- Human tissue, body parts, or blood/body fluids in a solid state can not be allowed to enter a storm drain. Appropriate BMPs should be implemented to prevent tissue from entering or being washed into the storm drain system; tissue found in the storm drain system should be removed to the maximum extent practicable.

- Dispose of sharps waste in a rigid sharps container. Dispose of the sealed container in an appropriate manner in compliance with the Medical Waste Management Act.

- The Orange County Health Care Agency (HCA), Regulatory Health Services may be contacted if needed for consultation/assistance in trauma scene management or questions regarding legal disposition of medical waste.

- For incidents on freeways, discuss with CHP the potential use of

CALTRANS for cleanup of trauma scene waste.

- When a Registered Trauma Scene Management Practitioner (RTSMP) is called to a scene, each agency or jurisdiction shall have a policy in place that establishes which department (fire, law enforcement, public works, etc.) is responsible for staying on scene until the RTSMP arrives on scene. In no case shall the scene be abandoned prior to the arrival of the RTSMP.

Specific Cleanup Procedure for Small Spills:

Public Agency personnel may decontaminate and then wash down small amounts of human blood/body fluids that are still in a liquid state. A “small” spill shall be defined as a spill that can be thoroughly and completely disinfected with one gallon of 10% chlorine disinfectant. This determination should be made with the understanding that multiple applications may be necessary to accomplish full disinfection.

When it is determined that the incident may be designated as small the following procedures shall apply:

- Apparatus shall carry a Hudson sprayer, maximum capacity 1 gallon. The sprayer shall be filled to 90% capacity with clean tap water.

- In a separate container, apparatus shall carry liquid household chlorine bleach and the ability to measure ¼ cup quantities of it.

- When decontamination of a liquid trauma scene waste becomes necessary, personnel shall add a minimum of ¼ cup of the liquid bleach to the water in the Hudson sprayer. This will create a 10% solution.

  - If an agency chooses to premix a 10:1 water to chlorine solution, they must replace it with fresh solution daily.

  - In lieu of using a 10% chlorine solution for decontamination, a Public Agency may follow a pre-described sanitization protocol outlined by OSHA (29 CFR 1910.1030) or other professionally recognized state or federal documents

- Personnel shall wear PPE’s to protect skin, mucus membranes, lungs and clothing from chlorine or trauma scene waste during the mixing and cleanup process.

- Personnel shall apply disinfectant to the liquid biowaste (they may see foam develop) and let it sit for a minimum of 10 minutes.

- After 10 minutes, personnel shall wash down the area using a fire hose, pressurized water extinguisher, etc.

- In non-emergency situations, and when circumstances permit, personnel should:
    - Attempt to move trash, or miscellaneous debris from the path of the runoff so no additional or unnecessary material is discharged to be carried into the storm drain.

    If the incident location provides personnel with a choice between directing the runoff down a storm drain, or a sewer, always use the sewer (with permission from the sewering agency). o Porous surfaces such as asphalt may require multiple repetitions of the procedure to completely disinfect and thoroughly clean the area. Unless the repeat decontamination and wash down procedure can be accomplished until the area is thoroughly cleaned from start to finish with 1 gallon of 10% bleach solution, a RTSMP should be called.

    - No biowaste in a solid state, or the water associated with the clean up of this waste shall be washed down a storm drain. If present, call a RTSMP to handle the cleanup.

  If the material requiring cleanup meets any of the following criteria, Public Agency personnel shall call in a Registered Trauma Scene Management Practitioner (RTSMP).

- Criteria:
    - The amount of <u>liquid</u> trauma scene waste at the incident reasonably exceeds the ability to thoroughly disinfect the incident with 1 (one) gallon of 10% chorine disinfectant.
    - The material requiring clean up is human tissue or a body part, and Coroner's Office personnel will not be responding to the scene to remove the material because the incident did not involve a fatality.

- A Registered Trauma Scene Management Practitioner (RTSMP), can be requested through dispatch when needed.

- When a Registered Trauma Scene Management Practitioner (RTSMP)

is called to a scene, each agency or jurisdiction shall have a policy in place that establishes which department (fire, law enforcement, public works, etc.) is responsible for staying on scene until the RTSMP arrives on scene. In no case shall the scene be abandoned prior to the arrival of the RTSMP.

C. Cleanup of Trauma Scene Wastes on Private Property

- The property owner has two options. Hire a Registered Trauma Scene Management Practitioner (RTSMP) or cleanup the affected area personally while observing the following restrictions.
    - Employees of the property owner who have not been given OSHA mandated Blood borne Pathogens training, proper personal protective equipment, and adequate supplies of bleach and water to complete the cleanup operation shall not be used for this task.
    - Trauma scene wastes must be disposed of properly and shall not be discharged to a storm drain.
    - A list of RTSMP's that offer trauma scene waste cleanup service may be obtained from the California Department of Public Health web page at http://www.cdph.ca.gov/certlic/medicalwaste/Documents/MedicalWaste/2013/Practitioners.pdf. **Note**: List is not a recommendation for these companies.

D. General Personnel and Clothing Clean-Up/Decontamination

After a potential exposure to bodily fluids proper decontamination of personnel and clothing may include the following:

- Hand washing
- Wear gloves when washing or handling contaminated equipment, clothing or other materials. Wear other personal protective gear as necessary according to potential for splashing.
    - With Water - Immediately after contact with blood or bodily fluids, or immediately after removing gloves, wash hands well with soap and running water in an area that is connected to a sanitary sewer.
    - Without Water - When running water and soap are not available, use a waterless hand wash substitute until personnel are able to

wash hands appropriately.

- Clothing decontamination: Boots, either leather or rubber, may be cleaned with germicide and allowed to air dry. Uniforms contaminated in the field with potentially infectious materials, blood, or other bodily fluid should be removed on scene. The outer shirt may be removed and placed in a labeled plastic bag or bio-hazardous linen bag. Pants may be removed, placed in a labeled plastic bag or bio-hazardous linen bag, and replaced with a Tyvex suit or a cover gown from an exposure kit to protect vehicle from contamination while returning to station.

- Contaminated uniforms should be fully cleaned by laundering according to procedure after return to station. Spot cleaning is not appropriate under any circumstances.

2. Discharges Associated with Emergency Fire Fighting Activities

To the extent allowed by the circumstances at the scene and without compromising the health and safety of personnel or the public, emergency fire fighting activities should be performed in a manner that avoids or minimizes discharges to the storm water system. BMPs that may be considered during emergency fire fighting activities include the following:

- If possible, avoid directing fire fighting flows directly on erodible surfaces if runoff will enter receiving waters or storm drains;

- If possible, apply fire-fighting flows such that runoff will flow over vegetated areas prior to entering receiving waters or storm drains.

3. Discharges Associated with Hazardous Materials Spills

The Orange County Health Care Agency (HCA), Environmental Health Division is the certified unified program agency with countywide responsibility for coordinating response to hazardous material spills (24–hour emergency response through Control 1 at (714) 628-7008). Fire departments within the County are participating agencies with specified responsibilities within their respective jurisdictions. Each department operates under a Hazardous Materials Area Plan that describes procedures for the allocation of resources and assigns tasks in time of a hazardous materials emergency. Fire department and safety personnel are trained to respond to hazardous material spills according to response protocols established by each department under oversight of the HCA. BMPs for hazardous materials emergencies are set forth in the current response protocols for each department.

4. Notification of Spills, Releases and Illegal Discharges

Spills, releases, and illegal discharges of pollutants to the receiving waters or to the storm drains shall be reported by the Discharger as required by all applicable state and federal laws. In addition, any such spills, releases and illegal discharges with the potential to endanger health, safety or the environment shall be reported by fire department staff to County of Orange Public Works within 24-hours after discovery of the spill, release or discharge. If safe to do so, necessary actions shall be taken to contain and minimize the spill, release or illegal discharge.

IMPLEMENTATION STRATEGY

Education, Training, and Outreach

In addition to the annual Stormwater Awareness Training that is provided to the Fire Prevention Officers and Fire Chiefs Association by the Principal Permittee (DAMP Section 10.9.5), fire department personnel within the County will receive targeted education and outreach at local fire stations to increase staff awareness and understanding of stormwater pollution issues as well as their compliance obligations. The Permittees plan to implement two levels of staff education and training – General and Activity Specific.

1. General Stormwater Training

All fire department personnel will receive basic introduction to stormwater pollution issues by means of General Stormwater Workshops created and funded by the Stormwater Program. While created by Stormwater Program staff, the workshops typically will be given by fire department trainers to the staffs of their respective departments. A “Train the Trainer” workshop will be created and given by the Stormwater Program to familiarize fire department trainers with material and subject matter prior to implementation of General Stormwater Workshops for fire department personnel. The following educational materials may be used in the General Stormwater Workshops:

- BMP booklets, fact sheets and other materials prepared for public dissemination;
- General stormwater training video;
- Employee Knowledge and Behavior Survey – given before and after each General
- Stormwater Workshop.

Through these efforts, fire department staff is also reminded about the Storm Water Hot Line phone number and web page. Additional materials may become available as cooperative efforts at the watershed and regional level are expanded.

2. Activity Specific BMP Training

Fire departments that perform activities identified in this Manual will fund, create, and conduct Activity Specific training sessions to introduce personnel to new work procedures, functions, and behaviors that incorporate the BMPs recommended in this manual. Fire departments will work closely with the Stormwater Program staff to create a complete training module for personnel and to establish a system to update and improve the information and training materials available to staff. The following Activity Specific training activities may be conducted by fire departments:

- Activity Specific Workshop – information and demonstration will be provided to ensure all personnel are familiar with the proper methods of performing BMPs applicable to their activities. New employees will receive the training during their orientation program;

- Tailboard Training Session – each department will conduct Tailboard Training Sessions appropriate for their work groups. Training sessions will address all topics related to BMPs applicable to the workgroups;

- Stormwater BMP Bulletin Board – will be provided in personnel areas of each station and will contain a listing of BMPs applicable to the station personnel.

- Activity Specific Reference Binders – will be developed and maintained at each station. Binders will contain training materials for each Tailboard Training Session, along with training records and other training materials from the Stormwater Program

3. Manual/BMP Update

The Permittees will continue to work cooperatively with fire departments to identify, update, and provide guidance on the implementation BMPs as appropriate to reduce contaminants in discharges related to fire agency activities to the maximum extent practicable.

PROGRAM ASSESSMENT AND REPORT

Program Effectiveness Assessment Strategy

The Permittees will assess effectiveness of the program described in this manual at least annually by implementing the following assessment procedures:

- Document all education and training activities conducted by Stormwater Program staff and fire department staff,
- Quantify fire department staff receiving educational materials and training (see Section 4);
- Inspect a selected number of fire stations to assess compliance with recommended BMPs;
- Develop and conduct a survey of Fire Prevention Officers, Fire Chiefs Association, and contacts at individual fire stations to obtain feedback on practicality and effectiveness of BMPs recommended in this manual and to obtain suggestions for additional BMP's and for revising currently listed BMPs.

Annual Report

Activities performed by the Permittees under this stormwater program element, results of any assessment surveys and inspections, and any revisions made to this manual will be documented annually in the Permitees' Annual Report.